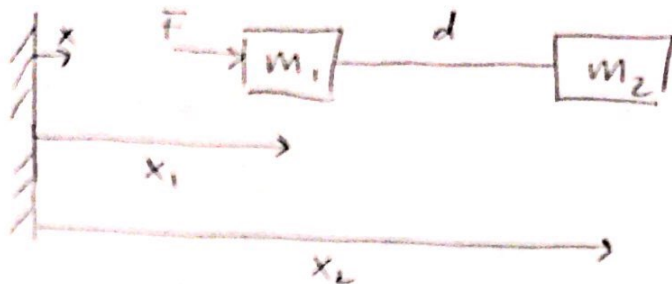


example

1D



constrained system

Goal:

Find $\ddot{x}_1$ and $\ddot{x}_2$

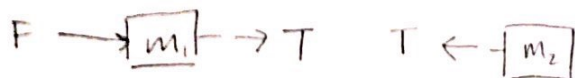
Given $x_1, x_2, \dot{x}_1, \dot{x}_2$

Ans: $\ddot{x}_1 = \ddot{x}_2 = \frac{F}{(m_1 + m_2)}$

Compare w/ example * :



FBD for both ex , and ex^*



True for both systems

LMB:

$$F + T = m_1 \ddot{x}_1 \quad -T = m_2 \ddot{x}_2$$

For ex^* ,

$$T = k \left(\frac{x_2 - x_1 - l_0}{l} \right)$$

For ex

$$T = ?$$

- cannot find T from $x_1, x_2, \dot{x}_1, \dot{x}_2$

BIG Q: How to solve constrained problems?

What is a constrained problem?

Have system with no constraints and n degrees of freedom

Add m constraints and have $n - m$ DoF

Number of minimal coordinates = # dof < # of maximal coordinates

How to find $\frac{d^2}{dt^2}$ of confiscated variables?

Various Methods:

- 1) Minimal coordinates using various theorems + tricks w/ LMB, AMB, Power
- 2) Various methods based on "fundamental equation of dynamics"
 - a) Lagrange equations
 - b) Gibbs - Appell
 - c) Hamilton's equations
 - d) Kane's eqns
 - e) ...
 - f) ...
- 3) Maximal coordinates
 - use N-E (Newton-Euler)
 - add constraint equations, set up + solve DAE's: differential algebraic equations
 - a) High order (ode45, etc)
 - b) First order (impulse based) ← NOT do in this class

Same example:

Method 1: LMB for system

1 DoF $\Rightarrow$ use x_1 as minimal coordinate

$$x_G = x_1 + \underbrace{\left(\frac{m_2}{m_1 + m_2} \right) d}_\text{constant}$$
$$\ddot{x}_G = \ddot{x}_1 \rightarrow \text{constant}$$

$$\left\{ \sum \vec{F} = \dot{\vec{L}} \right\} \cdot \hat{i}$$

$$F = \underbrace{(m_1 + m_2)}_{m_{tot}} \ddot{x}_1$$

$\uparrow$
 x_G

$$\ddot{x}_1 = \frac{F}{m_1 + m_2}$$

Method 2:

Minimal coordinates

$$P = \dot{E}_k$$

$$Fv = \frac{d}{dt} \left(\underbrace{\frac{1}{2} (m_1 + m_2) \dot{x}_1^2}_{E_k} \right) + \cancel{E_{k16}}^0$$

$$F \dot{x}_1 = (m_1 + m_2) \dot{x}_1 \dot{x}_1$$

- Assume $\dot{x}_1 \neq 0$

$$F = (m_1 + m_2) \ddot{x}_1 \Rightarrow \boxed{\ddot{x}_1 = \frac{F}{(m_1 + m_2)}}$$

Method 3: Lagrange Equations- Pick minimal coordinate: $q = x_1$ - Calculate $E_k(q, \dot{q}) \Rightarrow E_k = \frac{(m_1 + m_2)}{2} \dot{x}_1^2$ - Calculate $E_p(q) \Rightarrow E_p = 0$ - Calculate Q - that thing such that $Q dq = dW$
 $\uparrow$
 dx_1

$$Q = F \Leftarrow F dx_1 = dW$$

- Write Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = Q$$

$$\boxed{\mathcal{L} = E_k - E_p}$$

$$\frac{d}{dt} \left(\frac{\partial}{\partial \dot{x}_1} \left(\frac{1}{2} (m_1 + m_2) \dot{x}_1^2 \right) \right) - \frac{\partial}{\partial x_1} \left(\frac{1}{2} (m_1 + m_2) \dot{x}_1^2 \right) = F$$

$$\frac{d}{dt} ((m_1 + m_2) \dot{x}_1) - 0 = F$$

$$(m_1 + m_2) \ddot{x}_1 = F$$

$$\boxed{\ddot{x}_1 = \frac{F}{(m_1 + m_2)}}$$

Method 4: Maximal coordinates $\Leftrightarrow$ DAE

$\vec{F} = m\vec{a}$ for each particle:

$$F + T = m_1 \ddot{x}_1 \quad (1)$$

$$-T = m_2 \ddot{x}_2 \quad (2)$$

Constraint Equations:

$$x_1 + d = x_2$$

$\rightarrow$ differentiate twice

$$\ddot{x}_1 - \ddot{x}_2 = 0 \quad (3)$$

(1), (2), (3) are 3 equations + 3 unknowns
 $\ddot{x}_1, \ddot{x}_2, T$

Linear equations always

$$m_1 \ddot{x}_1$$

$$-1 \cdot T = F$$

$$m_2 \ddot{x}_2 + 1 \cdot T = 0$$

$$-\ddot{x}_1 + \ddot{x}_2 = 0$$

$$\underbrace{\begin{bmatrix} m_1 & 0 & -1 \\ 0 & m_2 & -1 \\ -1 & 1 & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ T \end{bmatrix}}_z = \underbrace{\begin{bmatrix} F \\ 0 \\ 0 \end{bmatrix}}_b$$

$$Az = b$$

Matlab:

$$A = \begin{bmatrix} m_1 & 0 & -1 \\ 0 & m_2 & -1 \\ -1 & 1 & 0 \end{bmatrix};$$

$$b = [F \ 0 \ 0]';$$

$$z = A \setminus b;$$

$$x_{idd} = z(1)$$